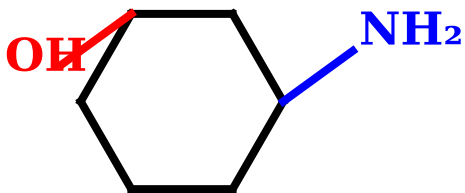
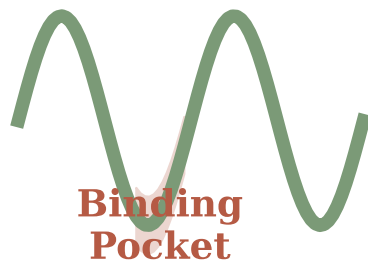


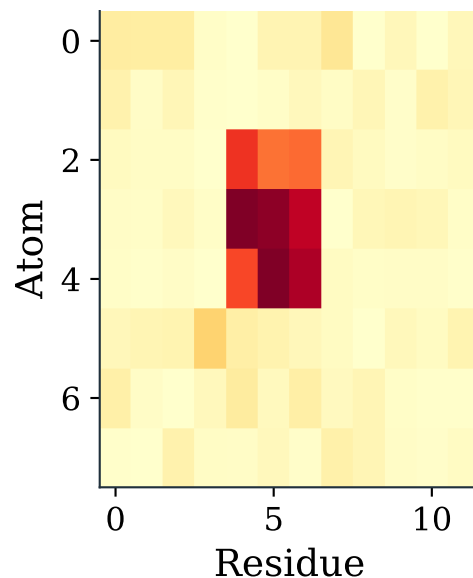
Drug (GINE GNN)



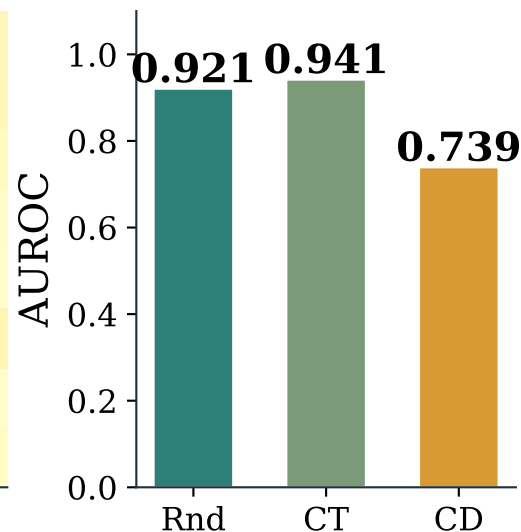
Protein (ESM-2 + Domain)



Cross-Attention Interaction Map



Performance



99.2% Attention Sparsity

Matches localised
binding-pocket focus

Conserved Attention Across Mutants

V104 remains dominant
across ABL1 variants

Pharmacophore Discovery

Amide > Amino >
Carbonyl > Halogen

Cold-Target Transfer

AUROC 0.941
through kinase homology